

Azure DevOps Organization Setup and Security Checklist

This comprehensive checklist guides IT administrators through the process of setting up a new Azure DevOps organization with proper security controls and permission management.

Pre-Setup Planning

1. Requirements Gathering

- ☐ **Business Requirements Analysis**
 - ☐ Identify number of expected users (Stakeholder/Basic/Visual Studio licenses needed)
 - ☐ Determine project structure and team organization
 - ☐ Define compliance requirements (SOX, GDPR, industry standards)
 - ☐ Establish security and governance policies
 - ☐ Plan integration requirements (Azure AD, third-party tools)
- ☐ **Technical Requirements**
 - ☐ Verify Azure subscription and billing setup
 - ☐ Confirm Azure AD tenant configuration
 - ☐ Plan network and firewall requirements
 - ☐ Determine backup and disaster recovery needs
 - ☐ Assess existing tool integrations
- ☐ **Stakeholder Alignment**
 - ☐ Get executive sponsorship and budget approval
 - ☐ Define roles and responsibilities
 - ☐ Establish change management processes
 - ☐ Plan user training and adoption strategy
 - ☐ Set success metrics and KPIs

2. Architecture Planning

- ☐ **Organization Structure Design**
 - ☐ Single organization vs. multiple organizations decision
 - ☐ Project hierarchy and naming conventions
 - ☐ Team structure within projects
 - ☐ Service connection strategy
 - ☐ Agent pool architecture
- ☐ **Security Architecture**
 - ☐ Identity and access management strategy
 - ☐ Network security requirements
 - ☐ Data classification and protection
 - ☐ Audit and compliance framework
 - ☐ Incident response procedures

Organization Configuration

3. Initial Organization Setup

- ☐ **Create Azure DevOps Organization**
 - ☐ Navigate to <https://dev.azure.com>
 - ☐ Sign in with organizational account
 - ☐ Create new organization with appropriate name
 - ☐ Select region for data residency compliance
 - ☐ Configure billing and licensing
- ☐ **Basic Organization Settings**
 - ☐ Set organization display name and description
 - ☐ Configure time zone and locale settings
 - ☐ Set up organization-level policies
 - ☐ Configure external guest access policies
 - ☐ Enable audit logging

4. Azure AD Integration

- ☐ **Connect to Azure Active Directory**
 - ☐ Link Azure DevOps organization to Azure AD tenant
 - ☐ Verify domain ownership
 - ☐ Configure user synchronization
 - ☐ Test authentication flow
 - ☐ Set up conditional access policies
- ☐ **Group Rules Configuration**
 - ☐ Create Azure AD groups for different roles
 - ☐ Configure Azure DevOps group rules
 - ☐ Set up automatic license assignment
 - ☐ Test group membership synchronization
 - ☐ Document group mapping strategy

5. Security Policies and Settings

- ☐ **Organization-Level Security Policies**
 - ☐ Configure third-party application access policies
 - ☐ Set up IP address restrictions if required
 - ☐ Enable/disable anonymous access to projects
 - ☐ Configure external guest access settings
 - ☐ Set up SSH key policies
- ☐ **Audit and Compliance Settings**
 - ☐ Enable audit logging
 - ☐ Configure audit log retention
 - ☐ Set up audit log streaming (if required)
 - ☐ Configure compliance dashboard
 - ☐ Document audit procedures

Project Setup Workflow

6. Project Creation and Configuration

- ☐ **Create Initial Projects**
 - ☐ Create projects based on planned structure
 - ☐ Configure project visibility (private/public)
 - ☐ Set project description and documentation
 - ☐ Configure project-level policies
 - ☐ Set up project dashboards
- ☐ **Project Security Configuration**
 - ☐ Review default security groups
 - ☐ Create custom security groups as needed
 - ☐ Configure project-level permissions
 - ☐ Set up area and iteration paths
 - ☐ Configure team settings

7. Repository Setup and Security

- ☐ **Repository Configuration**
 - ☐ Create initial repositories
 - ☐ Set up repository structure and naming conventions
 - ☐ Configure default branch (main/master)
 - ☐ Set up .gitignore and repository templates
 - ☐ Configure repository policies
- ☐ **Branch Protection Policies**
 - ☐ Protect main/master branches
 - ☐ Configure minimum reviewer requirements (minimum 2 reviewers)
 - ☐ Enable comment resolution requirements
 - ☐ Set up build validation policies
 - ☐ Configure automatic reviewer assignment
 - ☐ Restrict force push and delete permissions

8. Pipeline Security and Configuration

- ☐ **Build Pipeline Security**
 - ☐ Create secure agent pools
 - ☐ Configure pipeline permissions
 - ☐ Set up service connections with minimal permissions
 - ☐ Configure variable groups and secrets management
 - ☐ Implement pipeline security scanning
- ☐ **Release Pipeline Security**
 - ☐ Configure deployment environments
 - ☐ Set up environment approvals and checks
 - ☐ Configure release pipeline permissions
 - ☐ Implement deployment gates and validations
 - ☐ Set up monitoring and alerting

Security Group Configuration

9. Organization-Level Security Groups

- ☐ **Project Collection Administrators**
 - ☐ Limit membership to 2-3 essential personnel
 - ☐ Document members and their roles
 - ☐ Set up emergency access procedures
 - ☐ Configure MFA requirements
 - ☐ Establish regular review schedule
- ☐ **Custom Organization Groups**
 - ☐ Create security reviewers group
 - ☐ Set up audit readers group
 - ☐ Configure external contractor groups
 - ☐ Create service account groups
 - ☐ Document group purposes and permissions

10. Project-Level Security Groups

- ☐ **Standard Project Groups**
 - ☐ Configure Project Administrators (limit to 2-3 people)
 - ☐ Set up Contributors group with appropriate permissions
 - ☐ Configure Readers group for stakeholders
 - ☐ Create Build Administrators group
 - ☐ Set up Release Administrators group
- ☐ **Custom Project Groups**
 - ☐ Create role-specific groups (Developers, QA, etc.)
 - ☐ Set up external vendor groups
 - ☐ Configure temporary access groups
 - ☐ Create service-specific groups
 - ☐ Document group membership criteria

11. Team Configuration

- ☐ **Team Setup**
 - ☐ Create teams aligned with organizational structure
 - ☐ Configure team administrators
 - ☐ Set up team dashboards and boards
 - ☐ Configure team notification settings
 - ☐ Establish team working agreements

Permission Assignment Best Practices

12. User Access Management

- ☐ **User Onboarding Process**
 - ☐ Create standardized onboarding checklist
 - ☐ Assign appropriate access levels (Stakeholder/Basic/Visual Studio)

- ☐ Add users to correct Azure AD groups
- ☐ Provide security training and documentation
- ☐ Set up access review schedule
- ☐ **Service Account Management**
 - ☐ Create dedicated service accounts for automation
 - ☐ Use managed identities where possible
 - ☐ Implement service principal authentication
 - ☐ Document service account purposes
 - ☐ Set up credential rotation schedule

13. Permission Validation

- ☐ **Access Testing**
 - ☐ Test user access with different permission levels
 - ☐ Validate service connection permissions
 - ☐ Test branch policy enforcement
 - ☐ Verify pipeline security controls
 - ☐ Validate audit logging functionality
- ☐ **Security Validation**
 - ☐ Perform penetration testing (if required)
 - ☐ Validate MFA enforcement
 - ☐ Test conditional access policies
 - ☐ Verify backup and recovery procedures
 - ☐ Conduct security review with stakeholders

Post-Setup Validation Steps

14. Operational Readiness

- ☐ **Documentation and Training**
 - ☐ Create user guides and documentation
 - ☐ Conduct administrator training
 - ☐ Provide end-user training sessions
 - ☐ Document support procedures
 - ☐ Create troubleshooting guides
- ☐ **Monitoring and Alerting**
 - ☐ Set up security monitoring dashboards
 - ☐ Configure alerting for security events
 - ☐ Implement usage monitoring
 - ☐ Set up performance monitoring
 - ☐ Create compliance reporting

15. Ongoing Maintenance

- ☐ **Regular Review Processes**
 - ☐ Schedule monthly permission reviews
 - ☐ Plan quarterly security audits

- ☐ Set up annual access certifications
- ☐ Create change management procedures
- ☐ Establish incident response procedures
- ☐ **Automation and Optimization**
 - ☐ Implement automated permission auditing
 - ☐ Set up automated user provisioning
 - ☐ Create self-service access request processes
 - ☐ Implement automated compliance reporting
 - ☐ Optimize performance and costs

Compliance and Governance

16. Audit and Compliance Setup

- ☐ **Audit Configuration**
 - ☐ Enable comprehensive audit logging
 - ☐ Configure audit log retention policies
 - ☐ Set up audit log analysis and reporting
 - ☐ Create compliance dashboards
 - ☐ Document audit procedures
- ☐ **Governance Framework**
 - ☐ Establish governance committee
 - ☐ Create policy and procedure documentation
 - ☐ Set up regular governance reviews
 - ☐ Implement change control processes
 - ☐ Create risk management procedures

17. Business Continuity

- ☐ **Backup and Recovery**
 - ☐ Document backup procedures
 - ☐ Test recovery procedures
 - ☐ Create disaster recovery plan
 - ☐ Set up alternative access methods
 - ☐ Train backup administrators
- ☐ **Knowledge Management**
 - ☐ Create administrator runbooks
 - ☐ Document all configurations
 - ☐ Maintain contact lists and escalation procedures
 - ☐ Create knowledge transfer procedures
 - ☐ Set up regular knowledge sharing sessions

Final Validation Checklist

18. Go-Live Readiness

- ☐ **Technical Validation**

- ☐ All systems tested and functional
- ☐ Security controls validated
- ☐ Performance benchmarks met
- ☐ Backup and recovery tested
- ☐ Integration points validated
- ☐ **Organizational Readiness**
 - ☐ Users trained and ready
 - ☐ Support procedures in place
 - ☐ Documentation complete
 - ☐ Stakeholder sign-off obtained
 - ☐ Communication plan executed

19. Post-Implementation

- ☐ **Immediate Actions (First Week)**
 - ☐ Monitor system performance and usage
 - ☐ Address any immediate issues
 - ☐ Collect user feedback
 - ☐ Validate security controls
 - ☐ Update documentation as needed
- ☐ **Short-term Actions (First Month)**
 - ☐ Conduct first permission audit
 - ☐ Review and optimize configurations
 - ☐ Address user training gaps
 - ☐ Implement feedback improvements
 - ☐ Plan next phase enhancements

Emergency Procedures

20. Break-Glass Access

- ☐ **Emergency Access Procedures**
 - ☐ Document emergency access accounts
 - ☐ Create break-glass procedures
 - ☐ Test emergency access regularly
 - ☐ Train emergency response team
 - ☐ Set up emergency communication channels

21. Incident Response

- ☐ **Security Incident Procedures**
 - ☐ Create incident response plan
 - ☐ Define escalation procedures
 - ☐ Set up incident tracking system
 - ☐ Train incident response team
 - ☐ Conduct regular incident response drills

Checklist Summary

Total Items: 150+ validation points across 21 categories

Estimated Timeline: 4-8 weeks for complete implementation (varies by organization size)

Key Success Factors: - Executive sponsorship and clear governance - Proper planning and requirements gathering - Phased implementation with validation at each step - Comprehensive training and documentation - Regular review and continuous improvement

Critical Security Controls: - Multi-factor authentication enforcement - Principle of least privilege implementation - Comprehensive audit logging - Regular permission reviews - Incident response capabilities

This checklist should be customized based on your organization's specific requirements, compliance needs, and risk tolerance. Regular updates to this checklist are recommended as Azure DevOps features and security best practices evolve.